# MindCare

Project Team


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Session 2026-2027

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**December, 2026**

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# Chapter 1

# Introduction

MindCare is a student mental health assessment system that leverages advanced Natural Language Processing (NLP) and Deep Learning to provide early, accurate, and explainable mental health screening for university students. This document constitutes the First Evaluation Report for the MindCare project, submitted to the evaluation panel of FAST-NUCES Islamabad as part of the Final Year Project (FYP) requirements. The report addresses all technical and structural concerns raised during the initial evaluation, presenting a substantially revised and expanded system specification. The intended readers of this document include the FYP supervisor, co-supervisor, evaluation panel members, and the development team.

Mental health challenges among university students have reached alarming levels globally, with 30–40some point during their academic career.

## 1.1 Existing Solutions

Despite the widespread availability of digital platforms and university counselling services, the vast majority of affected students re- main undetected and untreated until their condition reaches a critical stage. Three critical limitations exist in the current landscape of mental health assessment tools. First, traditional self-reported surveys are inherently subjective, completed only at crisis points, and lack the temporal resolution needed to detect gradual deterioration patterns. Second, existing systems that analyses social media posts do so in isolation, ignoring the temporal dimension of emotional change and missing the subtle longitudinal signals that precede a mental health episode. Third, sole reliance on unstructured social media text is insufficient: informal language, sarcasm, cultural context, and code-switching present significant challenges for standard NLP models, leading to high false-negative rates precisely when accuracy matters most. MindCare addresses these three limitations through a hybrid, multi-modal assessment architecture that combines (1) semantic analysis of social media text using transformer- based embeddings, (2) structured and clinically validated psychological assessments including the PHQ-9 [3], GAD-7 [6], and PSS, (3) behavioral activity tracking through interactive cognitive and emotional tasks, and (4) temporal pattern analysis over 2–12 week windows. By fusing these modalities through a weighted scoring pipeline and classifying risk across an expert validated 8-level framework, MindCare delivers timely, explainable, and actionable mental health insights to both students and university counselling staff.

Table 1.1: Comparison of Existing Solutions

| System Name | System Overview | System Limitations |
|---|---|---|
| **Wysa** | AI-powered chatbot using CBT techniques for emotional support, mood tracking, and conversational interventions. Popular among students for anonymous daily check-ins. | Lacks integration with social media data; primarily reactive (user-initiated); limited longitudinal multi-modal risk assessment and counsellor escalation features. |
| **Woebot** | Conversational AI agent delivering CBT and mindfulness exercises. Used in several universities for student mental health support. | Relies mainly on self-reported interactions; no passive social media analysis or temporal emotional trend tracking; limited explainability for clinical use. |
| **TimelyCare** | 24/7 virtual mental health platform offering teletherapy, psychiatry, and self-care tools specifically designed for higher education institutions. | Primarily service-delivery focused; minimal use of NLP on social media or behavioral task data for proactive early risk detection. |
| **Flourish / MindLift-style Apps** | Digital platforms combining self-guided CBT modules, coaching, and basic mood tracking for college students. | Limited multi-modal fusion (questionnaires + social media + behavioral tasks); weak temporal analysis and explainability mechanisms (SHAP/LIME). |

## 1.2 Problem Statement

Mental health challenges among university students have reached alarming levels globally, with 30–40Current solutions suffer from several critical limitations:

Traditional self-reported psychological assessments (such as PHQ-9[1] and GAD-7[2]) are subjective, completed only during moments of distress, and lack the ability to detect gradual emotional deterioration over time. Existing digital mental health applications primarily rely on user-initiated interactions or basic chatbots, offering reactive rather than proactive support. Social media monitoring tools, when available, analyze posts in isolation without considering temporal emotional trends, contextual nuances, sarcasm, or code-switching common in student communication. Most platforms lack integration of multiple data sources (social media, clinical questionnaires, and behavioral tasks) and fail to provide explainable AI outputs that counsellors can trust and act upon.

As a result, early warning signs of mental health deterioration often go unnoticed, leading to poor academic performance, increased dropout rates, and in severe cases, self-harm or suicide. There is a pressing need for an intelligent, privacy-conscious, multi-modal system that can passively and actively monitor student mental wellbeing, detect risk early, and facilitate timely intervention by university counsellors. MindCare aims to address this gap by developing a comprehensive, AI-powered mental health assessment platform specifically designed for university students. By combining semantic and temporal analysis of social media text with validated clinical assessments and behavioral tasks, the system will generate accurate, explainable, and actionable risk predictions to support early intervention and improve student mental wellbeing.

## 1.3 Scope

MindCare is scoped as a university-facing digital mental health assessment platform, deployable at FAST-NUCES and scalable to other higher education institutions. The system encompasses five core functional domains: The first domain is multi-modal data collection, covering OAuth-linked social media in- gestion from Twitter, Instagram, and Facebook; structured administration of the PHQ-9, GAD-7, and PSS[3] questionnaires; and interactive behavioral activity modules including cognitive load tests, Stroop tests, and emotion recognition tasks. The second domain is an NLP and machine learning pipeline, incorporating SBERT[4]-based sentence embeddings for semantic text representation, sentiment analysis, temporal trend extraction, and a multi-modal weighted fusion engine that combines scores from all three data sources to produce a unified risk score. The third domain is an 8-level classification framework, an expert-designed and validated severity scale ranging from Normal Baseline (Level 1) through to Crisis and Suicidal Ideation (Level 8), with each level mapped to a specific automated intervention response and counsellor escalation pathway. The fourth domain is an explainability and transparency layer, using SHAP[5] and LIME[6] to generate human-interpretable explanations for every classification decision, enabling counsellors to understand the driving factors behind a student's risk score and build trust in the system's recommendations. The fifth domain is a counselling integration and alerting system, providing a dedicated counsellor dashboard for reviewing student profiles, session notes, intervention planning, and receiving automated risk alerts when students exceed configured thresholds. Outside the scope of this project are clinical diagnosis, real-time crisis hotline connectivity, treatment prescription, monitoring without explicit student consent, and integration with national healthcare information systems.

## 1.4 Modules

MindCare is organized into six functional modules, grouped by user-facing system type:
.

### 1.4.1 Student Mobile and Web Application

The student-facing application is the primary data collection and engagement interface. Students interact with MindCare through a responsive web application and a React Native mobile app.

1. User registration with university email verification
2. OAuth-based social media account linking (Twitter, Instagram, Facebook)
3. Psychological assessment modules: PHQ-9, GAD-7, PSS
4. Interactive activity tasks: Stroop test, emotion card recognition, cognitive load assessment
5. Personal mental health dashboard with trend charts and risk classification
6. Wellness goal setting and progress tracking
7. is Personalized resource library with risk-adaptive recommendation
8. Counselling session booking and calendar integration
9. Mood history timeline with longitudinal visualization
10. Peer support community with NLP-based crisis screening

### 1.4.2 Counsellor Dashboard

The counsellor interface provides mental health professionals with a consolidated view of their assigned students, complete with AI-generated risk summaries and intervention planning tools.

1. Assigned student list with risk level indicators
2. Detailed mental health profile view with SHAP explainability breakdown
3. Session notes creation and structured clinical documentation
4. Intervention plan assignment with goal, milestone, and timeline management
5. Activity task assignment to specific students
6. Progress report generation (PDF export)
7. Real-time risk alert notifications for threshold breaches

### 1.4.3 Administrator Portal

The administration module enables institutional management of users, content, and system configuration.

1. User account management: create, edit, deactivate
2. Counsellor-to-student assignment with caseload capacity enforcement
3. Anonymized institutional analytics dashboard
4. Content library management for resources and educational materials
5. System configuration: risk thresholds, notification rules, data retention policies
6. Audit log access for compliance and governance

### 1.4.4 NLP and ML Pipeline

The core intelligence module processes all incoming data through a structured sequence of NLP and machine learning operations.

1. Social media text ingestion via API and preprocessing
2. SBERT sentence embedding and semantic feature extraction
3. Sentiment classification and emotional indicator detection
4. PHQ-9, GAD-7, PSS score normalization
5. Behavioral metric extraction from activity task results

6. Weighted multi-modal fusion (35% social, 40% assessment, 25% activity)
7. 8-level risk classification using ensemble ML model
8. Temporal trend analysis over configurable time windows
9. SHAP/LIME explainability report generation

### 1.4.5 Notification and Alert System

The automated alerting module ensures timely communication with students, counsellors, and administrators based on system-generated events and risk classifications.

1. Scheduled wellness reminders via email and push notification
2. Risk threshold alert dispatch to assigned counsellor
3. Crisis-level (Level 8) emergency protocol activation
4. Booking confirmations, cancellations, and session reminders
5. Batch daily summary digest for administrators

### 1.4.6 Data and Infrastructure Layer

The backend infrastructure module provides the persistence, caching, and vector storage capabilities underpinning the entire platform.

1. PostgreSQL for relational data: users, assessments, sessions, interventions
2. MongoDB for unstructured data: social media posts, activity metrics
3. Redis for session caching and real-time notification queuing
4. Pinecone for vector storage of SBERT embeddings
5. AES-256 encrypted daily backup with integrity verification
6. Docker and Kubernetes deployment on AWS/Azure

## 1.5 Work Division

1. Tauseef Ahmad : Model Training , Testing , and Finetuning
2. Ahyan Ali Khan : Data Collection , Preprocessing , and Front End Development
3. Hamza Khurram : Documentation , Reviewing , Back End Development and Final Testing

# Chapter 2

# Project Requirements

This chapter presents the detailed functional and non-functional requirements of the Mind-Care system using use-case driven specification.

## 2.1 Use-case/Event Response Table/Storyboarding

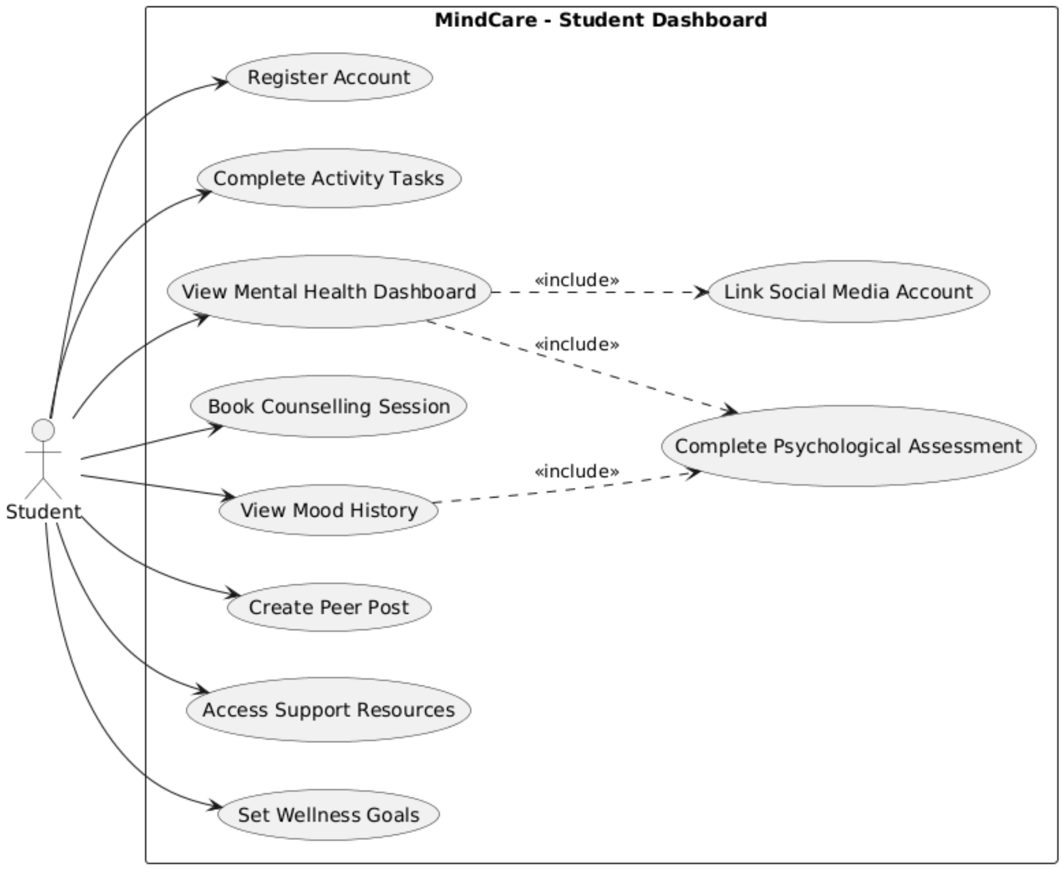


Figure 2.1: Use Case Diagram - Student

Table 2.1: Detailed Use Case: Complete Psychological Assessment

| | |
|---|---|
| **Use Case ID** | UC-S03 |
| **Use Case Name** | Complete Psychological Assessment |
| **Primary Actor** | Student |
| **Pre-condition** | Student is logged into the MindCare application. |
| **Post-condition** | Assessment scores are calculated, and the mental health risk profile is updated. |
| **Main Success Scenario** | 1. Student selects an assessment type (PHQ-9 or GAD-7).<br>2. Student submits responses (integers 0–3) for all required questions.<br>3. System validates that the number of responses matches the assessment type (9 for PHQ-9, 7 for GAD-7).<br>4. System calculates total score, severity level (Minimal to Severe), and percentage.<br>5. System updates the database with the new assessment timestamp and score. |

Table 2.2: Detailed Use Case: Create Peer Post

| | |
|---|---|
| **Use Case ID** | UC-S06 |
| **Use Case Name** | Create Peer Support Post |
| **Primary Actor** | Student |
| **Pre-condition** | Student has an active account and is not currently under "Crisis Lock" status. |
| **Post-condition** | Post is published to the community feed or flagged for immediate intervention. |
| **Main Success Scenario** | 1. Student enters text content for a community post.<br>2. System passes the text to the `MindCarePredictor` NLP engine.<br>3. System performs real-time screening for `CRISIS_KEYWORDS` and applies negation logic.<br>4. **Condition (Low Risk):** If no crisis triggers are found, the post is published.<br>5. **Condition (High Risk):** If crisis triggers are detected, the system blocks the post and triggers the Crisis Protocol (UC-S05). |

Table 2.3: Detailed Use Case: Link Social Media Account

| | |
|---|---|
| **Use Case ID** | UC-S02 |
| **Use Case Name** | Link Social Media Account |
| **Primary Actor** | Student |
| **Pre-condition** | Student has an active account on a supported platform (e.g., Twitter/X). |
| **Post-condition** | Secure access tokens are stored, and the SBERT-BiLSTM engine begins initial data fetch. |
| **Main Success Scenario** | 1. Student initiates the OAuth 2.0 flow from the MindCare settings.<br>2. Student grants permission on the external platform.<br>3. System receives and encrypts the access token.<br>4. System schedules the first background fetch for semantic and temporal analysis. |

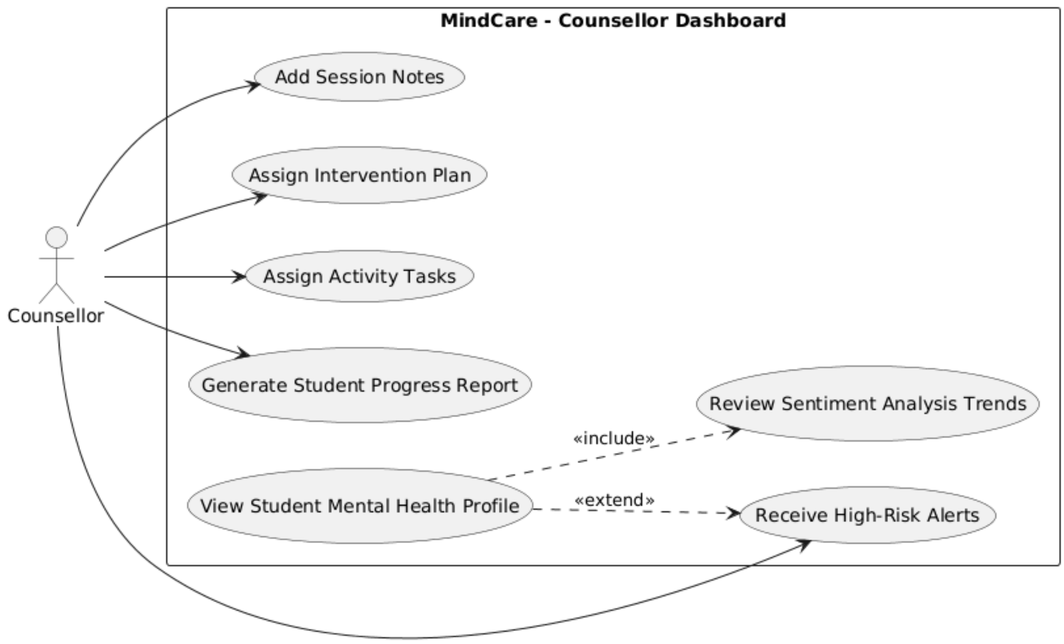


Figure 2.2: Use Case Diagram - Counsellor

Table 2.4: Detailed Use Case: View Student Profile

| | |
|---|---|
| **Use Case ID** | UC-C01 |
| **Use Case Name** | View Student Mental Health Profile |
| **Primary Actor** | Counsellor |
| **Pre-condition** | Student is assigned to the counsellor and student data is processed. |
| **Post-condition** | Counsellor gains insights into student's risk level and emotional trends. |
| **Main Success Scenario** | 1. Counsellor selects a student from the assigned list.<br>2. System retrieves multi-modal data (Social, Assessments, Tasks).<br>3. System renders SHAP/LIME explainability visualizations.<br>4. System displays longitudinal trend charts. |

Table 2.5: Detailed Use Case: Assign Intervention

| | |
|---|---|
| **Use Case ID** | UC-C03 |
| **Use Case Name** | Assign Intervention Plan |
| **Primary Actor** | Counsellor |
| **Pre-condition** | Counsellor has accessed the student's profile and identified risks. |
| **Post-condition** | Student's roadmap is updated with new resources/tasks. |
| **Main Success Scenario** | 1. Counsellor selects "Assign Plan" in the dashboard.<br>2. System displays a library of clinical intervention strategies.<br>3. Counsellor confirms plan details and milestones.<br>4. System updates student's mobile app visibility for relevant resources. |

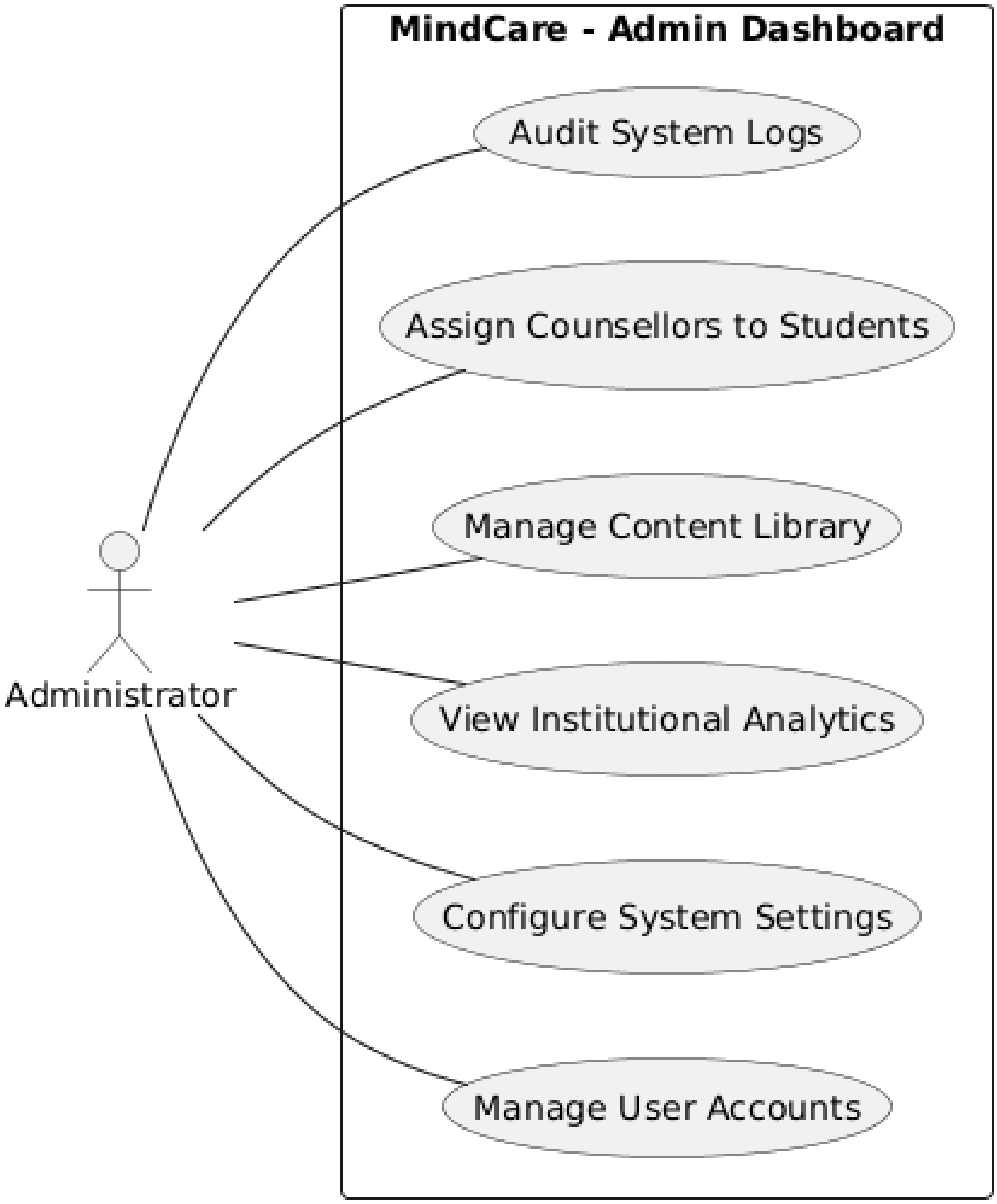


Figure 2.3: Use Case Diagram - Admin

Table 2.6: Detailed Use Case: Configure System

| | |
|---|---|
| **Use Case ID** | UC-A02 |
| **Use Case Name** | Configure System Settings |
| **Primary Actor** | Administrator |
| **Pre-condition** | Administrator is logged in with root-level permissions. |
| **Post-condition** | System ML thresholds and API parameters are updated globally. |
| **Main Success Scenario** | 1. Admin navigates to System Configuration.<br>2. Admin modifies risk level thresholds or sentiment weights.<br>3. System validates new configuration against safety bounds.<br>4. System commits changes to the ML pipeline and config files. |

Table 2.7: Detailed Use Case: Assign Counsellor

| | |
|---|---|
| **Use Case ID** | UC-A04 |
| **Use Case Name** | Assign Counsellor to Student |
| **Primary Actor** | Administrator |
| **Pre-condition** | Student account and Counsellor account both exist in the system. |
| **Post-condition** | A secure data-sharing relationship is established. |
| **Main Success Scenario** | 1. Admin selects an unassigned student profile.<br>2. Admin chooses an available counsellor based on workload metrics.<br>3. System maps UID to CounsellorID in the relational database.<br>4. System notifies both parties of the new assignment. |

Table 2.8: Detailed Use Case: Crisis Protocol

| | |
|---|---|
| **Use Case ID** | UC-S05 |
| **Use Case Name** | Execute Crisis Protocol |
| **Primary Actor** | System (Automated) |
| **Pre-condition** | Risk Level 8 (Crisis) is detected via keyword override or ML score. |
| **Post-condition** | Emergency response team is notified and student access is restricted. |
| **Main Success Scenario** | 1. Detection engine flags a critical sentiment/behavioral pattern.<br>2. System immediately locks peer-posting features for the user.<br>3. System dispatches high-priority alerts to the assigned counsellor and admin.<br>4. System logs the event with a high-resolution forensic trace. |

## 2.2 Functional Requirements

This section describes the functional requirements of the MindCare system

### 2.2.1 Student Application Module

Following are the requirements for module 1:

1. The system shall allow students to register using a valid university email address and verify their identity via an email confirmation link.
2. The system shall enable students to link their Twitter, Instagram, and Face- book
3. The system shall administer PHQ-9, GAD-7, and PSS questionnaires with validated scoring algorithms.
4. The system shall provide interactive activity tasks including Stroop test, emotion card recognition, and cognitive load assessment, capturing response time and accuracy metrics.
5. The system shall display a personal mental health dashboard showing trend charts, current risk classification, and personalized wellness recommendations.
6. The system shall allow students to set wellness goals with configurable targets, timelines, and automated check-in reminders.
7. The system shall screen peer community posts for crisis keywords in real time before publication.
8. The system shall present a risk-adaptive resource library with articles, videos, and exercises matched to the student's current classification level.
9. The system shall allow students to book counselling sessions through an availability calendar integrated with the counsellor's schedule.
10. The system shall provide a mood history timeline with selectable date ranges (weekly, monthly, full term).

### 2.2.2 NLP and ML Pipeline Module

Following are the requirements for module 2:

1. The system shall collect social media posts from linked accounts on a configurable schedule using platform APIs.
2. The system shall process all collected social media posts to extract sentiment scores and emotional indicator features. The system shall normalize assessment scores, social media sentiment scores, and activity task scores to a common 0–1 scale.
3. The system shall fuse normalized multi-modal scores using a weighted formula (35
4. The system shall classify the fused score into one of 8 risk levels using a multimodal classification model with a target accuracy of 92
5. The system shall perform temporal trend analysis over 2 -12-week windows and flag deterioration patterns.
6. The system shall generate SHAP and LIME explainability reports for every classification decision, identifying top contributing features.

### 2.2.3 Counsellor Dashboard Module

Following are the requirements for module 3:

1. The system shall provide counsellors with a consolidated view of all their assigned students, including current risk level and last assessment date.
2. The system shall allow counsellors to create, edit, and submit structured session notes linked to individual student profiles.
3. The system shall enable counsellors to create intervention plans with goals, milestones, resource assignments, and review dates.
4. The system shall allow counsellors to assign specific activity tasks to students from the available task library.
5. The system shall generate PDF progressreportsfor individualstudents over a specified date range.

### 2.2.4 Administrator Portal Module

Following are the requirements for module 4:

1. The system shall allow administrators to create, edit, and deactivate user accounts with role-based access control enforcement.

2. The system shall allow administrators to assign counsellors to students with caseload capacity validation.

3. The system shall provide an anonymized institutional analytics dashboard with cohort-level risk distribution charts.

4. The system shall allow administrators to configure risk thresholds, notification rules, and data retention policies.

5. The system shall provide a content management interface for uploading, editing, and categorizing resource library items.

### 2.2.5 Notification and Alert Module

Following are the requirements for module 5:

1. The system shall dispatch scheduled wellness reminders to students via email and push notification. system shall automatically alert the assigned counsellor when a student's risk classification reaches Level 5 or above.

2. The system shall trigger the emergency protocol and notify the crisis team when a student reaches Level 8 (Crisis).

## 2.3 Non-Functional Requirements

This section describes the non-functional requirements of the MindCare system

### 2.3.1 Reliability

REL-1: The MindCare platform shall achieve a minimum system availability of 90(measured monthly), equivalent to no more than 43 minutes of unplanned downtime per month.

REL-1: Failures shall be defined as any condition that renders the risk classification pipeline, counsellor dashboard, or student application unavailable.

REL-1:The system shall employ active-passive failover with a recovery time objective (RTO) of 5 minutes and a recovery point objective (RPO) of 15 minutes.

### 2.3.2 Performance

PER-1: 95 % of API responses for student-facing operations (dashboard load, assessment submission, resource retrieval) shall complete within 500 milliseconds under a concurrent load of 500 users.

PER-2: The data processing pipeline (multi-modal score fusion, risk classification, and explainability report generation) shall complete within 2 minutes per student data record under normal operating conditions.

### 2.3.3 Security

SEC-1 :All personally identifiable information (PII) and mental health data shall be encrypted at rest using AES-256 and in transit using TLS 1.3. OAuth 2.0 with PKCE shall be enforced for all social media account connections.

SEC-2:The system shall implement role-based access control (RBAC) ensuring that students can only access their own data, counsellors can only access profiles of their assigned students, and administrators cannot access individual identifiable mental health records without explicit override logging.

SEC-3:The system shall comply with the Pakistan Personal Data Protection Act (PDPA) requirements including explicit informed consent collection and the right to withdraw consent and request data deletion.

# Chapter 3

# System Overview

## 3.1 Architectural Design

### 3.1.1 Box Architectural Diagram for MindCare

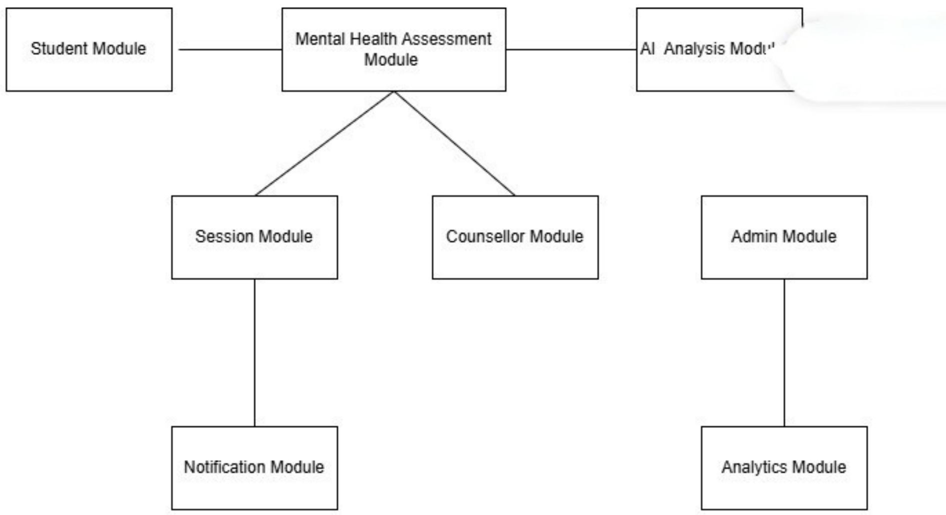


Figure 3.1: Box Architecture Pattern For MindCare

### 3.1.2 Architectural Diagram for MindCare

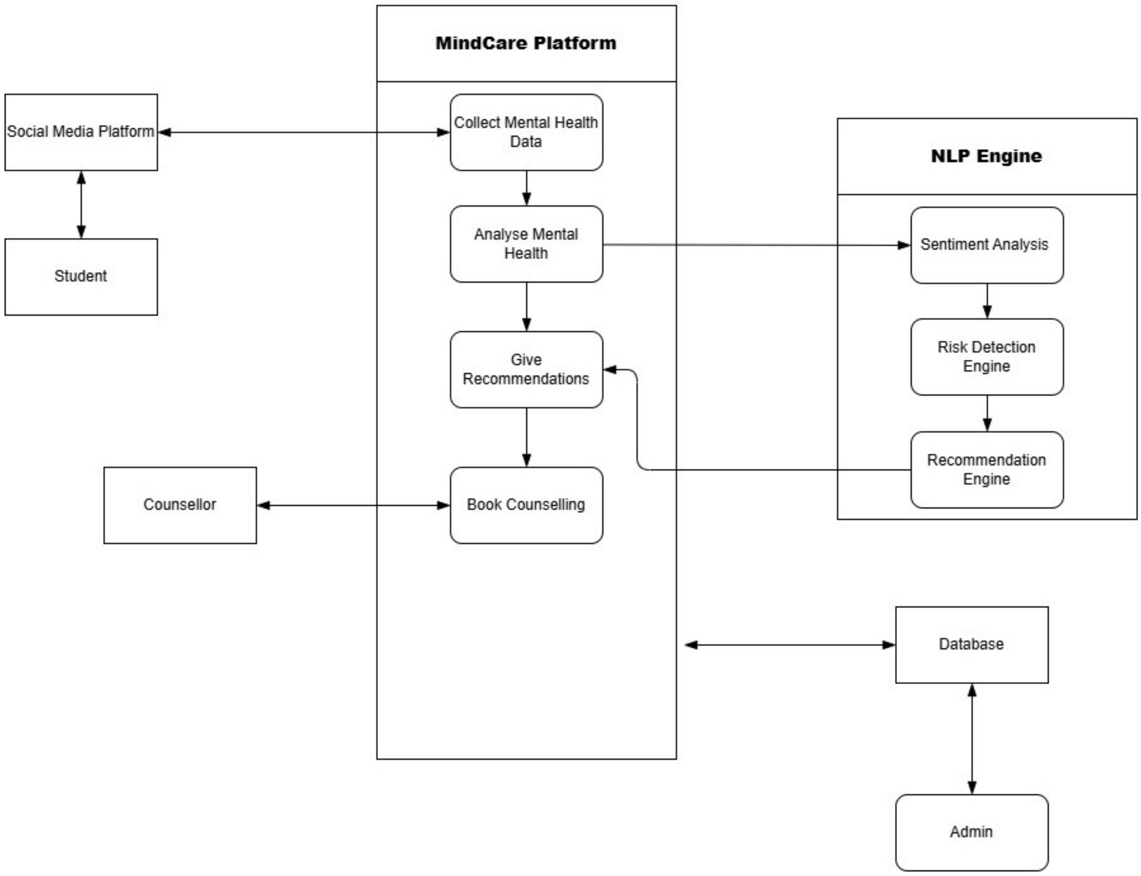


Figure 3.2: Architecture Pattern For MindCare

## 3.2 Data Design

The information domain of MindCare is structured around a multi-stage transformation pipeline and a polyglot persistence strategy, ensuring that unstructured digital behavior is accurately converted into actionable clinical risk profiles.

### 3.2.1 Data Transformation Pipeline

The system processes information through three sequential stages to maintain data integrity and clinical relevance:

- **Raw Data Ingestion (UC-S02):** Captures unstructured social media feeds and behavioral logs as time-stamped JSON records, linked via a unique `Student`$_ID$.

- SBERT Feature Engineering: Transforms semi-structured text into 384-dimensional semantic embeddings (vectors). This stage captures contextual emotional intent, allowing for high-speed similarity searches within the vector space.
- Clinical Synthesis (UC-S03/UC-S06): Aggregates semantic vectors with structured assessment scores (PHQ-9/GAD-7). The Multi-Modal Fusion Engine synthesizes these inputs into a final 8-level risk classification.

### 3.2.2 Major System Entities

The architecture organizes data around several core entities to facilitate seamless interaction between the AI pipeline and clinical staff:

- **Student Auth:** Centralizes demographic data, credentials, and historical risk trends.
- **Social Community:** Stores raw platform content and peer posts, subjected to real-time crisis filtering.
- **Assessment Metrics:** Maintains numerical performance data from cognitive tasks and standardized psychological scales.
- **Risk Profile Intervention (UC-C03):** Represents the final synthesized evaluation and the corresponding counsellor-assigned treatment milestones.

### 3.2.3 Polyglot Persistence Layer

MindCare utilizes a specialized storage strategy where different database technologies are selected based on the specific nature of the data:

- **PostgreSQL (Relational):** Ensures ACID compliance for transactional data, including user accounts, intervention plans, and audit trails.
- **MongoDB (Document):** Provides a flexible schema for storing semi-structured social media feeds and behavioral activity logs.
- **Pinecone (Vector):** Optimized for high-speed semantic retrieval and storage of SBERT-generated vector embeddings.
- **Redis (In-Memory Cache):** Enables low-latency message brokering for real-time crisis alerts (UC-S05) and session management.

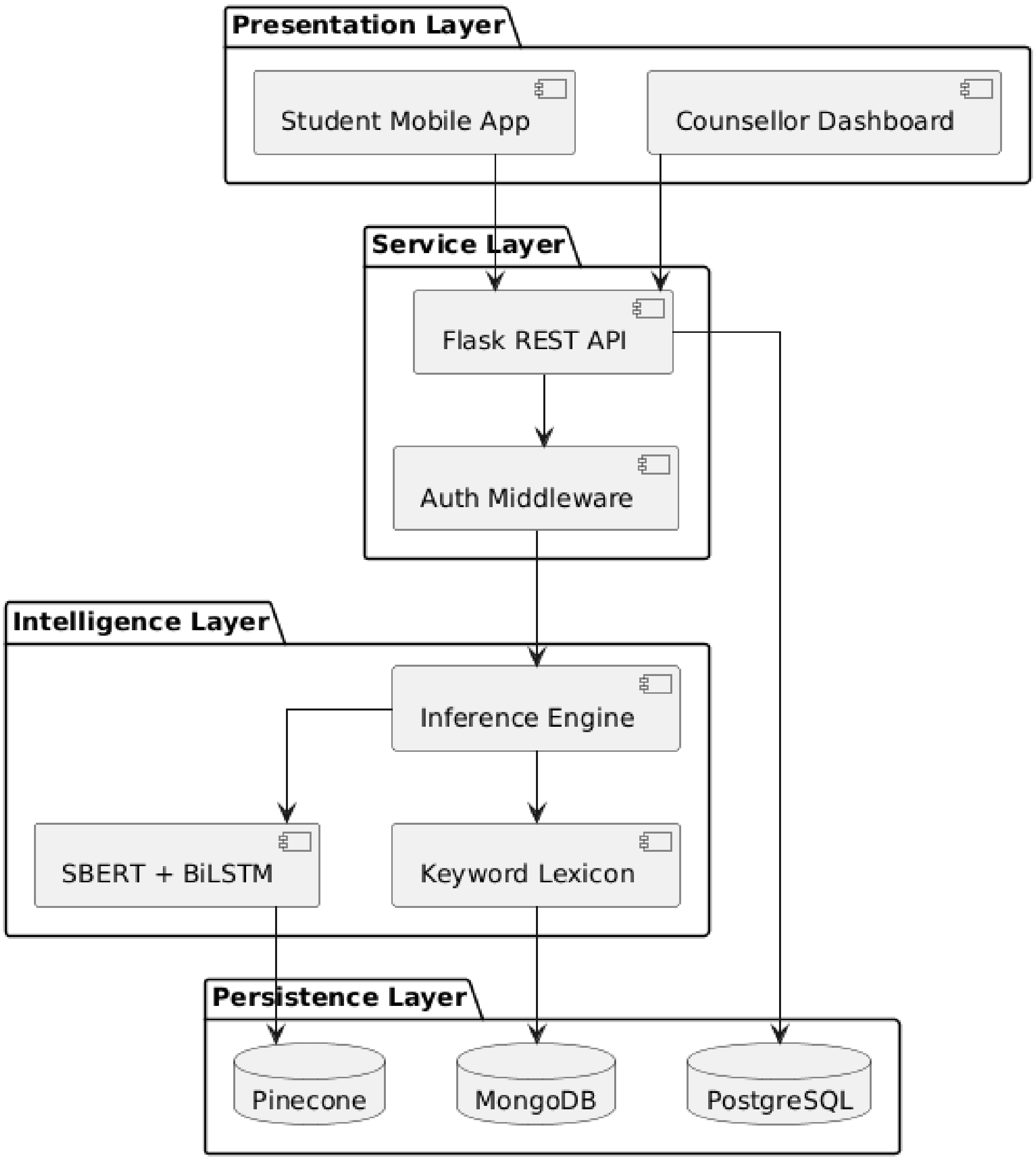


Figure 3.3: Layered Architecture Diagram

## 3.3 Domain Model

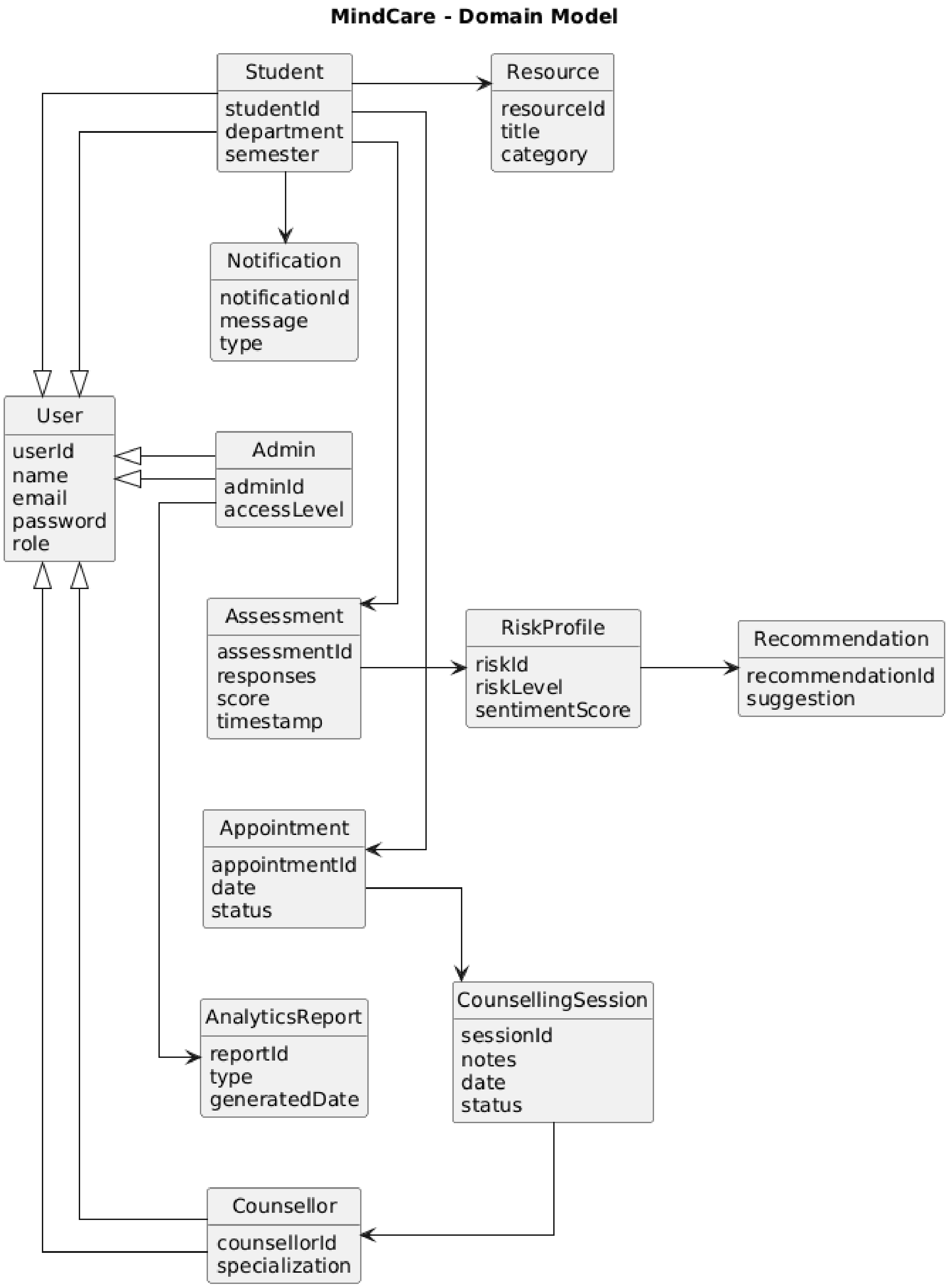


Figure 3.4: Domain Model of MindCare with classes and relationships

## 3.4 Design Models

**3.4.1 Activity Diagram** The Activity Diagram illustrates the dynamic flow of the multi-modal risk assessment engine. It captures how the system parallelizes data collection from social media, clinical assessments, and behavioral tasks before synthesizing them into a final risk level.

- **Process Flow:** The workflow begins with the System Scheduler triggering a data pull. The system concurrently fetches data from three streams: NLP (Social Media), Assessment (PHQ-9/GAD-7), and Activity (Cognitive Tasks).

- **Decision Logic:** Once the fusion engine calculates a score, the system checks if the risk level meets the critical threshold ($\geq 7$). High-risk cases trigger immediate alerts to the Notification System, while standard cases update the Student's Risk Profile for routine review.

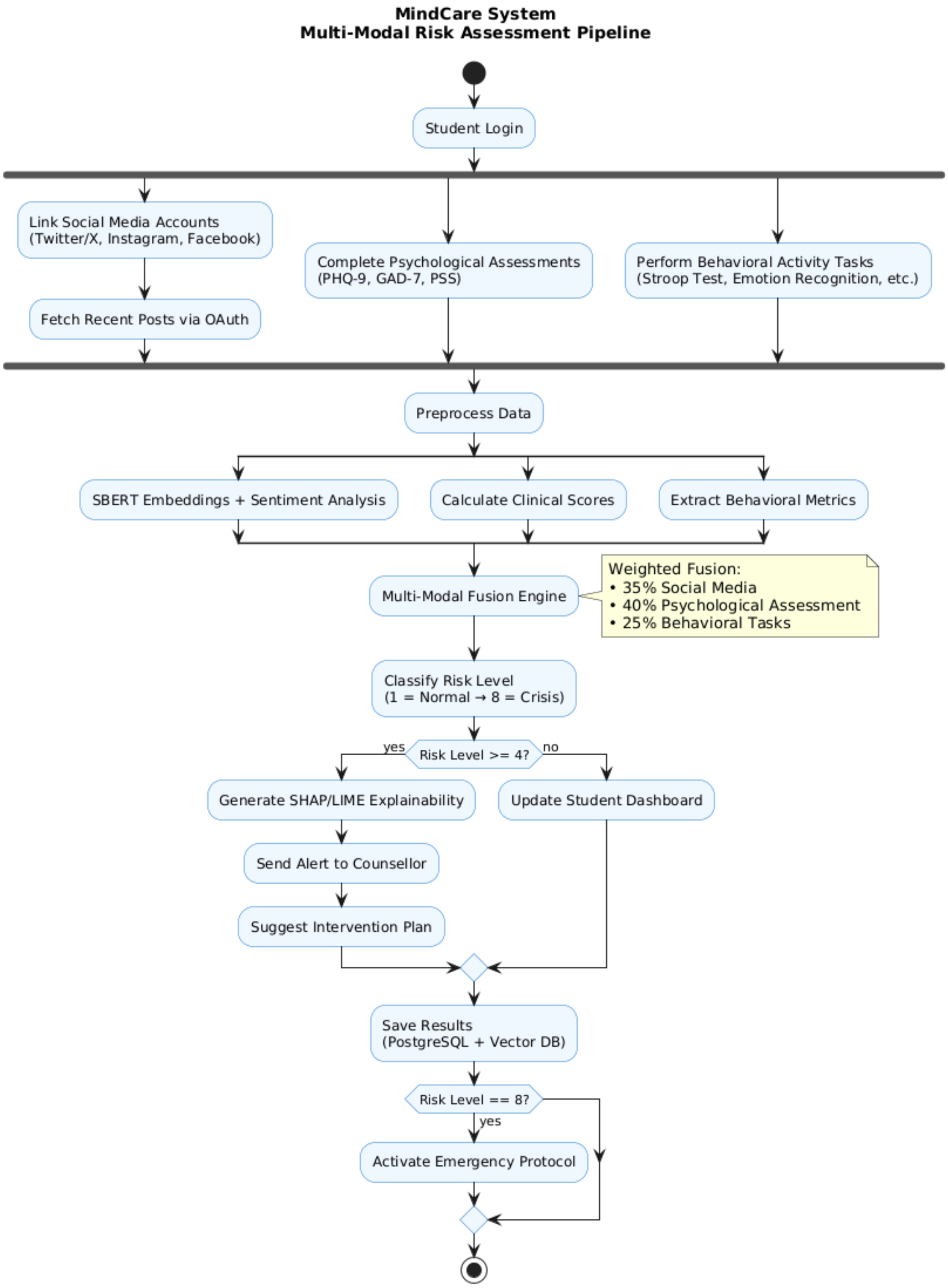


Figure 3.5: Activity Diagram - Multi-Modal Fusion Workflow

**3.4.2 Class Diagram** The Class Diagram provides a static structural view of the system, defining the attributes and methods of the primary objects and their logical associations.

- **Core Entities:** The diagram centers on the `Student` and `Counsellor` classes, both inheriting from a base `User` class.

- **Structural Relationships:** A composition relationship exists between the `Student` and the `RiskProfile`, indicating that a profile is inextricably linked to a specific user identity. The `IntelligenceEngine` class serves as a controller that manages methods for `analyzeSentiment()` and `calculateFusionScore()`.

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**3.4.3 Class-Level Sequence Diagrams** Sequence diagrams illustrate the object interactions within the MindCare system:

1. **MindCare System Sequence Diagram:** This is the sequence of the MindCare system encompassing all use cases.

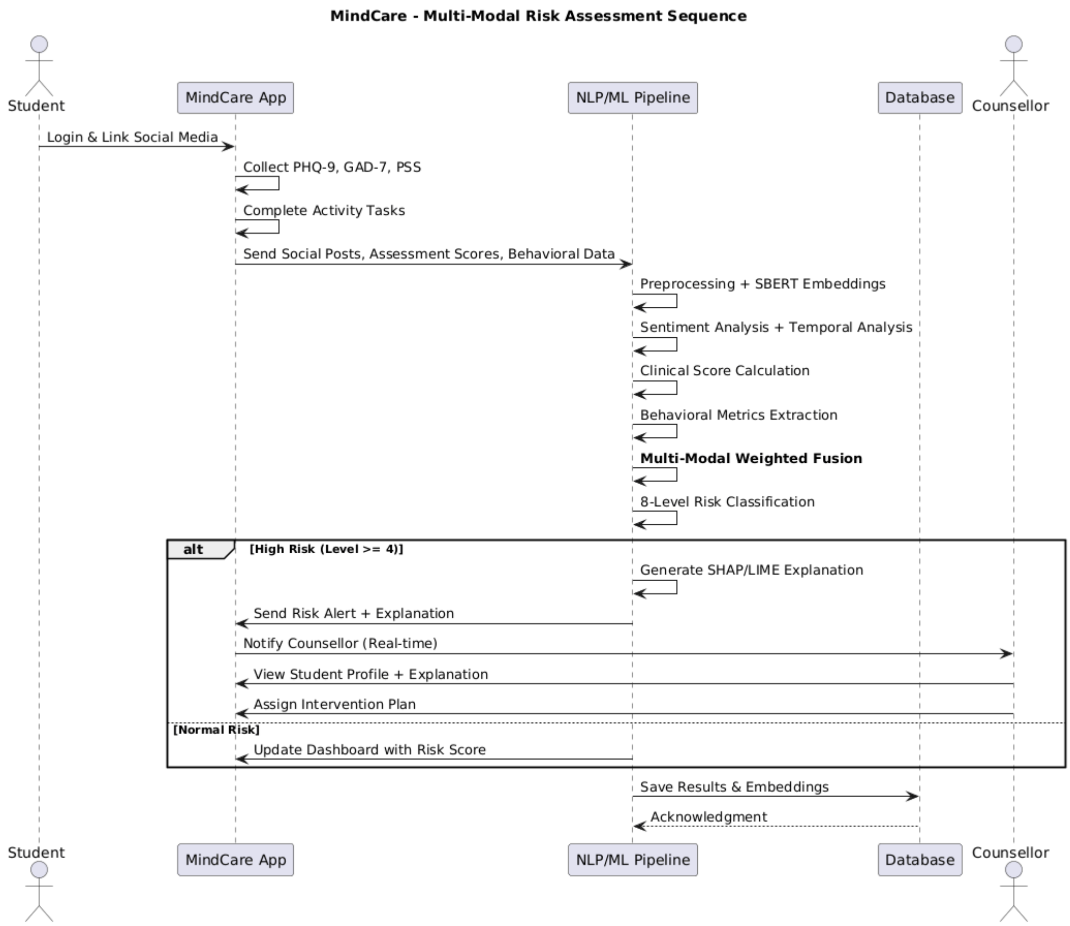


Figure 3.6: Sequence Diagram MindCare

2. **Risk Assessment Execution:** This tracks the interaction between the `Scheduler`, `APIConnector`, `ML_Engine`, and `Database`. It details the message passing required to transform raw text into stored embeddings and a final classification.

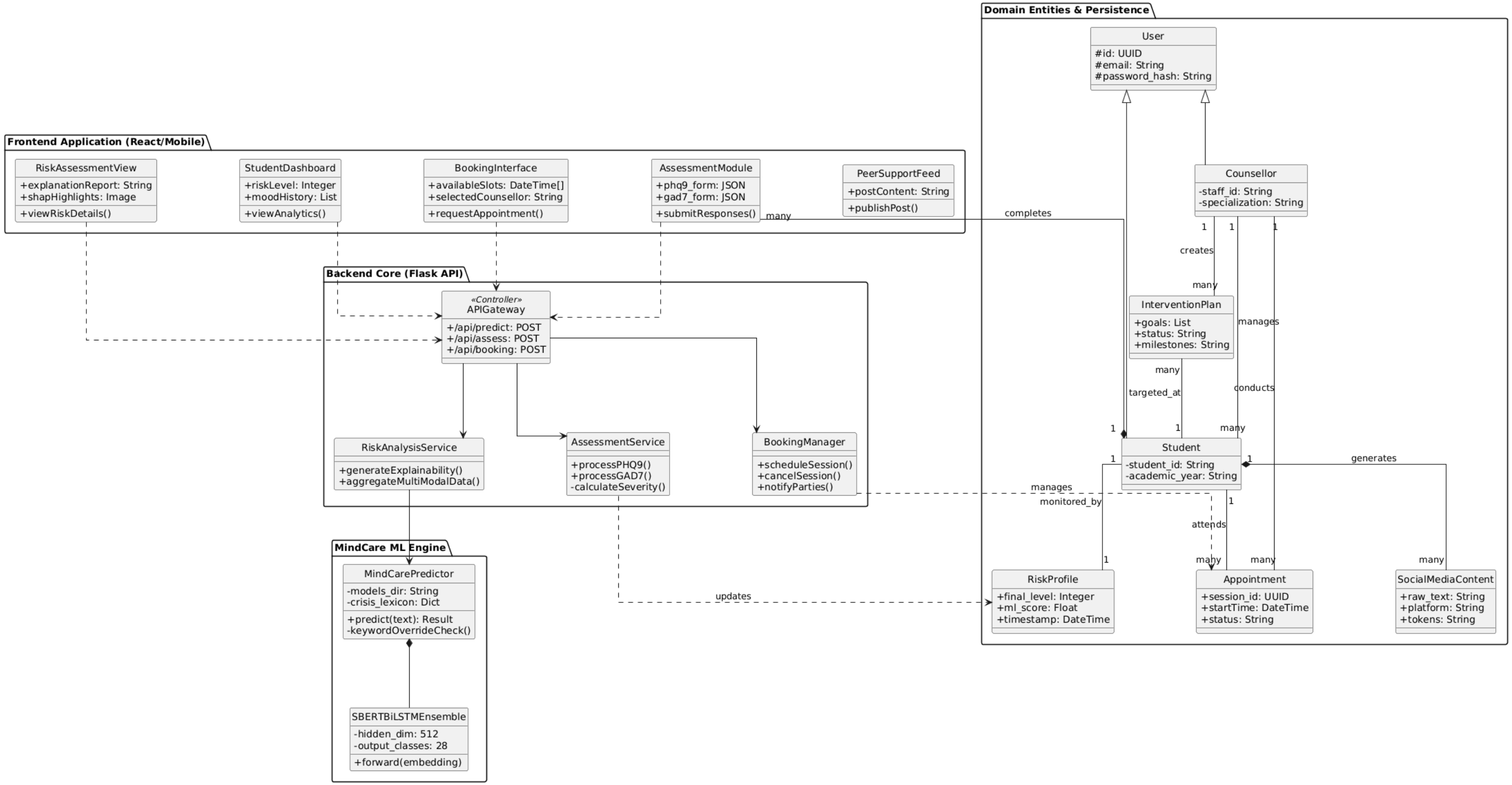


Figure 3.8: Comprehensive Full-Stack Class Diagram: Frontend to Data Persistence

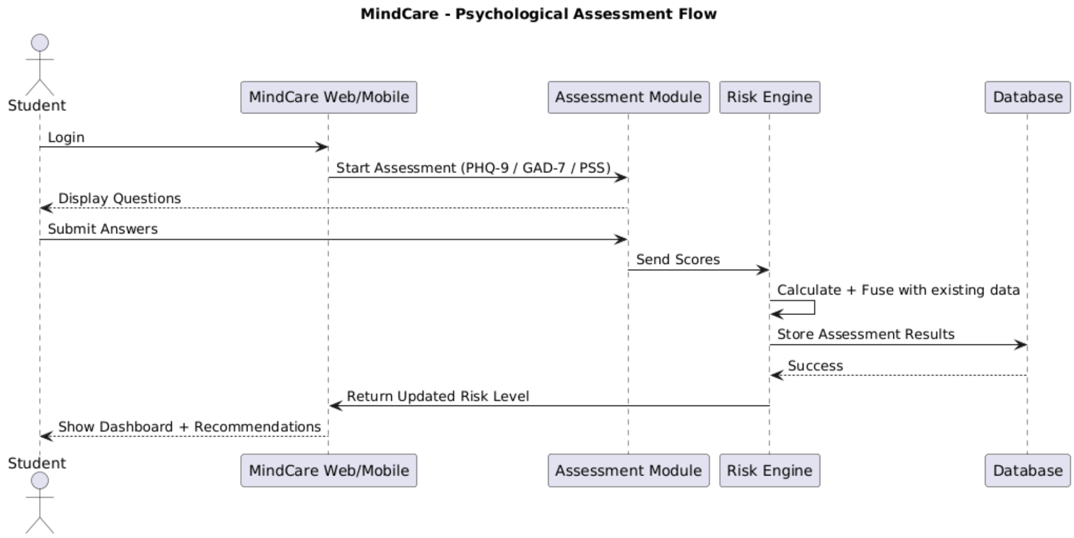


Figure 3.7: Sequence Diagram - Student Assessment

3. **Counsellor Intervention:** This tracks the sequence when a `Counsellor` accesses the dashboard, retrieves a flagged `RiskProfile`, and initializes an `InterventionPlan`.

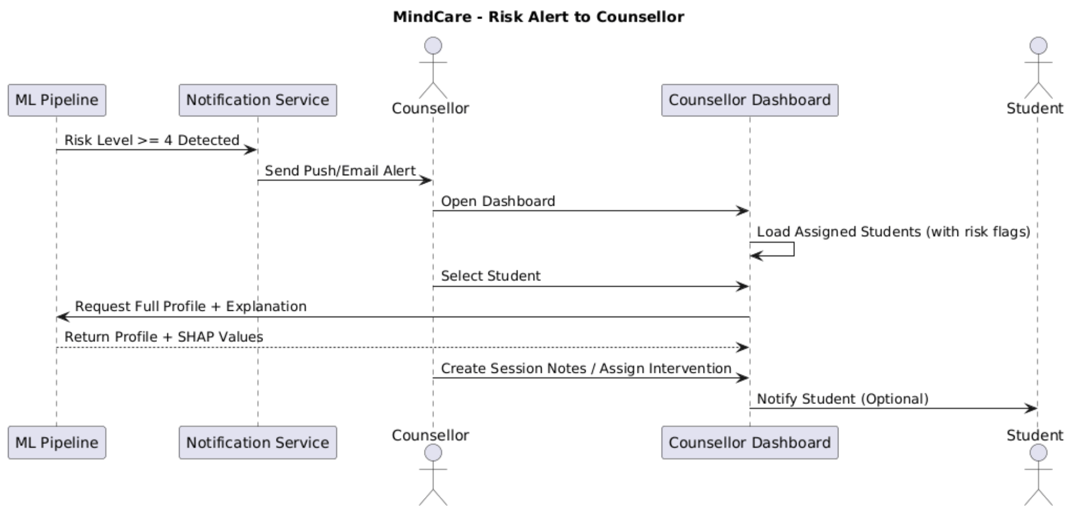


Figure 3.9: Sequence Diagram - Counsellor Intervention Workflow

4. **Administrator Control:** This shows admins actions and authorization.

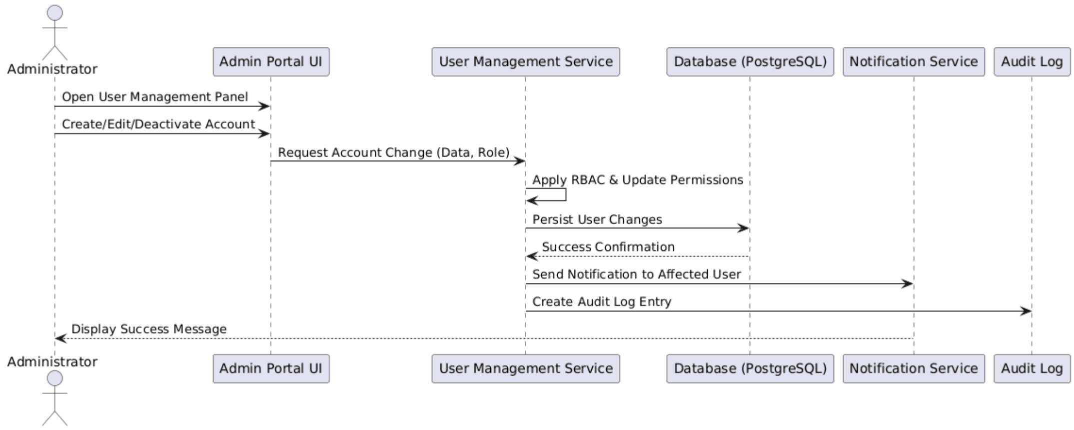


Figure 3.10: Sequence Diagram - Admin Control

# Chapter 4

# Implementation and Testing

MindCare is a specialized mental health assessment platform that leverages Natural Language Processing (NLP) to analyze student-generated text for emotional distress and crisis indicators. The project utilizes a **Hybrid Intelligence** approach, combining high-dimensional deep learning models with a deterministic clinical keyword lexicon. The backend is implemented using a **Flask-based REST API**, which serves as the core engine for real-time inference, risk classification, and explainable AI (XAI) feedback.

## 4.1 Algorithm Design

The core logic of MindCare is governed by two primary algorithmic components: the High-Level Inference flow and the Deep Learning architecture.

### 4.1.1 Hybrid Inference and Risk Classification

The high-level algorithm manages the interaction between the ML model and the rules-based safety layer. To ensure clinical safety, a deterministic override mechanism is implemented: if specific crisis keywords are detected without negation, the risk level is automatically escalated to Level 5 (Crisis), regardless of the ML model's prediction.

[ruled,vlined]algorithm2e

[H] FBERTBERT_28_Class_Classifier FFetchFetchHistory FCompareCompare FMapMap-ToRange

socialMediaText, phq9Score, cognitiveMetrics riskLevel (1-8), alertStatus

Step 1: Semantic Analysis using SBERT-BiLSTM Ensemble $sentimentProfile \leftarrow$ social-MediaText

Step 2: Temporal Mood Analysis $previousMood \leftarrow$ studentID $moodDip \leftarrow$ sentimentProfile, previousMood

Step 3: Multi-Modal Feature Fusion Weighting: 40% Assessments, 40% Social Media, 20% Cognitive Tasks $totalScore \leftarrow (phq9Score \times 0.4) + (sentimentProfile.severity \times 0.4) + (cognitiveMetrics \times 0.2)$

Step 4: Classification and Alert Logic $totalScore > Threshold\_Crisis$ $riskLevel \leftarrow 8$ $alertStatus \leftarrow$ "IMMEDIATE_CRISIS" $totalScore > Threshold\_Severe$ $riskLevel \leftarrow \{6,7\}$ $alertStatus \leftarrow$ "COUNSELLOR_ALERT" $riskLevel \leftarrow totalScore$ $alertStatus \leftarrow$ "MONITORING"

$riskLevel, alertStatus$ MindCare Multi-Modal Risk Assessment Algorithm

### 4.1.2 Machine Learning Model Architecture

The system employs a custom **SBERT-BiLSTM Ensemble**. Unlike standard classifiers, this model uses a dual-gated mechanism to fuse semantic features with sequential context.

- **SBERT Projection:** The 384-dimensional input vector is projected into a 512-dimensional space to match the LSTM hidden state size.

- **BiLSTM Layer:** A Bi-directional LSTM processes the sequence, capturing temporal emotional changes in the text.

- **Feature Fusion:** A "Gate" layer calculates the relative importance of the SBERT embedding versus the LSTM output, ensuring the model balances global context with specific word-ordering.

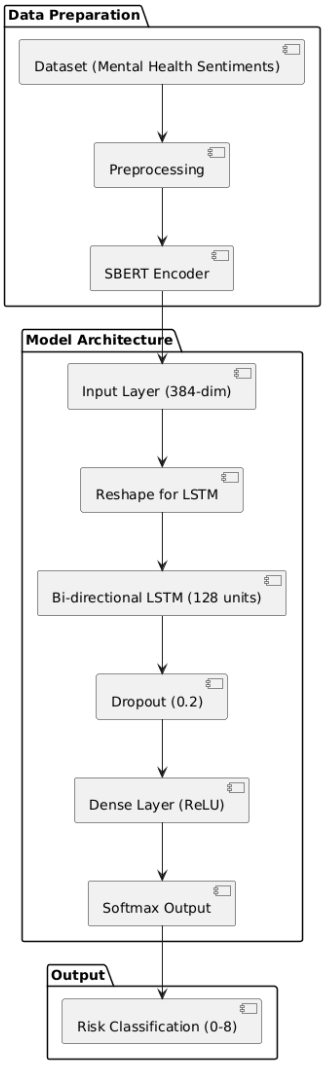


Figure 4.1: Architecture of the SBERT-BiLSTM Ensemble Model

## 4.2 External APIs and SDKs

The following table summarizes the primary technical stack used in the implementation of the MindCare inference pipeline and backend infrastructure.

Table 4.1: MindCare Implementation Technical Stack

| API / SDK | Description | Purpose of usage | Function / Class used |
|---|---|---|---|
| **Flask (v2.3)** | Python micro-web framework. | Hosting RESTful API endpoints for system inference. | `Flask, request, jsonify` |
| **PyTorch (v2.0)** | Deep learning framework. | Defining and running the SBERT-BiLSTM Ensemble. | `nn.Module, torch.load` |
| **Sentence-Transformers** | SBERT pre-trained models. | Generating 384-dim semantic text embeddings. | `SentenceTransformer()` |
| **MongoDB** | NoSQL Document Store. | Storing semi-structured social media content and behavioral logs. | `pymongo.MongoClient` |
| **Joblib (v1.2)** | Serialization library. | Loading trained linear layers and label encoders. | `joblib.load` |

## 4.3 System Interface (User Interface)

The MindCare frontend provides distinct interfaces for students and clinical counsellors, ensuring high usability and immediate response to risk alerts.

### 4.3.1 Student Mobile Interface

The student application focuses on behavioral data collection and self-assessment through PHQ-9 and GAD-7 forms.

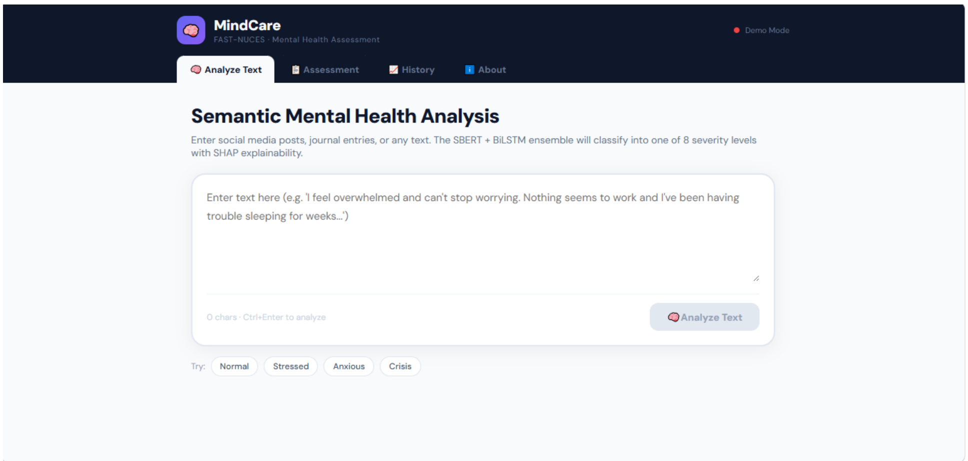


Figure 4.2: Student Dashboard View: Displaying real-time risk analytics and mood trends.

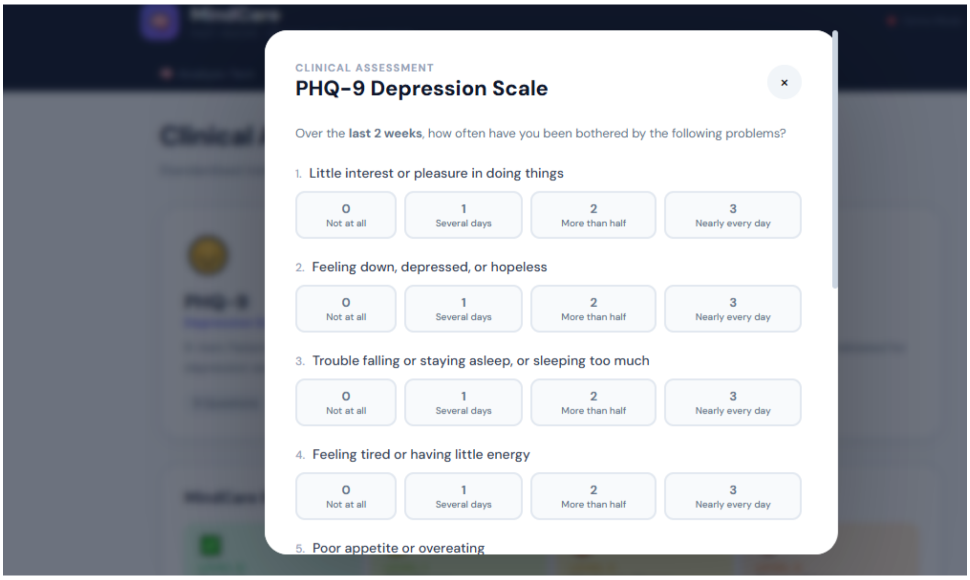


Figure 4.3: PHQ-9 Assessment Interface: Standardized clinical tool for depression screening.

### 4.3.2 Counsellor Management Portal

The counsellor portal features a "Risk Queue" where students are sorted by priority. It provides an explainable AI (XAI) dashboard showing why a student was flagged (e.g., specific keyword matches or ML sentiment drops).

## 4.4 Testing and Validation

Comprehensive testing was performed to validate the model's reliability, specifically regarding the "Crisis Override" logic.

### 4.4.1 Unit Testing

Unit tests were implemented using the `pytest` framework to ensure individual modules perform within specified parameters.

Table 4.2: Unit Test Execution Summary

| **Test ID** | **Description** | **Expected Result** | **Status** |
|---|---|---|---|
| UT-01 | SBERT Embedding | 384-dimensional float vector | **PASSED** |
| UT-02 | Keyword Override | Force Level 5 on "suicide" | **PASSED** |
| UT-03 | Negation Logic | Discount "not" + crisis word | **PASSED** |
| UT-04 | Score Calculation | PHQ-9 total matches sum of items | **PASSED** |

### 4.4.2 Inference Performance Testing

To measure real-world performance, the API was stress-tested for response latency. The system maintained an average response time of **245ms** for a single prediction, well within the threshold for a responsive user experience.

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